



A lightweight approach for network intrusion detection in industrial cyber-physical systems based on knowledge distillation and deep metric learning

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ARTICLE INFO

Keywords:

Intrusion detection
Industrial cyber-physical system
Knowledge distillation
Triplet neural network

ABSTRACT

With the rapid development of technology and science, machine learning approaches and deep learning methods have been widely applied in industrial Cyber-Physical Systems. However, there are still some challenging issues for anomaly detection to classify various attacks in industrial CPS to ensure the cyber security, especially when dealing with resource-constrained IoT devices. In this paper, we propose a Knowledge Distillation model based on Triplet Convolution Neural Network to improve the model performance and greatly enhance the speed of anomaly detection for industrial CPS as well as reduce the complexity of the model. Specifically, during the training process, we design a robust model loss function to improve the training stability of the model. A new neural network training method called K-fold cross training is also proposed to enhance the accuracy of anomaly detection. A lot of experimental results demonstrate that the performance metrics of KD-TCNN on the benchmark datasets NSL-KDD and CIC IDS2017 have significant advantages over traditional deep learning approaches and the recent state-of-the-art models. Furthermore, when compared to the original model, our model's computational cost and size are both reduced by roughly 86% with just 0.4% accuracy loss.

1. Introduction

A Cyber-Physical System (CPS) is a computer system made up of interacting parts with physical input and output that is monitored or controlled by computer algorithms. Through the industrial CPS, it can make the system more reliable, efficient and real-time collaboration, so it has gained a wide attention from governments as well as industry and has been widely applied. The industrial CPS far exceeds the existing industrial control network in terms of remote coordination capability, autonomy, type and number of control objects, and especially the scale of nodes for equipment within the system. It's universally acknowledged that the industrial CPS is characterized by geographically-dispersed, large-scale, sophisticated and heterogeneous. The development and implementation of different sorts of industrial CPS has increased exponentially in recent years, which have had a great impact on all part of our life, especially in the power grid, transportation systems, healthcare devices, home appliances (Dibaji, Pirani, Flamholz, Annaswamy,

Johansson, & Chakraborty, 2019; Luo, Guo, Yang, & Sun, 2019; Zhang, Qiu, Tsai, Hassan, & Alamri, 2017) and so on. A large number of these systems are deployed in important infrastructure, such as water supply equipment, wearable devices in healthcare, or in the places which are of importance to our daily lives. The range of CPS applications distributed throughout IoT networks, on the other hand, makes them easier to being exposed to cyber and physical threats, especially in the process of information transmission in intelligent manufacturing industry (Zhou, Liang, Shimizu, Ma, & Jin, 2021). These potential weaknesses introduce a security risk in the industrial CPS applications, causing the program to get out of control and harm the people who rely on it (Ding, Han, Xiang, Ge, & Zhang, 2018; Han, Xie, Chen, & Ling, 2014). Industrial CPS attaches great importance to communication and network capabilities, which collects data on the status of objects in the physical world in real time through networks and interfaces then sends it to the server which processes the data and then returns it to the physical devices to make the corresponding changes. Commonly, the CANbus network will firstly be

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attacked by the hackers, then they will hijack the data and sent the fake data to the server thereby compromising the equipment of the industrial CPS. The signals and information generated across the cyber network are scanned and gathered by the SCADA framework where deep learning models or machine learning approaches are applied to identify abnormal traffic (Zhou, 2021). Fig. 1 illustrates the entire industrial CPS architecture.

Intrusion detection systems (IDS) can detect intrusions that cannot be stopped by other security mechanisms, and it plays an important role in protecting the industrial CPS as a second line of defense. We can classify intrusion detection systems into Host-based Intrusion Detection System (HIDS) (Rebecca, 1998:) and Network-based Intrusion Detection System (NIDS) (Hamed, Dara, & Kremer, 2018) based on various data sources. HIDSs only monitor hosts, which need to be installed on each host, cannot observe and analyze network-related behavioral information. While NIDSs monitor numerous hosts as well as analyzes real-time traffic to gather packet information and evaluate its contents to detect abnormal traffic in the network (Chawla, Lee, & Fallon, 2019). Although a large number of models have been shown to yield good intrusion detection results in industrial CPS, identifying anomaly network traffic in this domain is still a challenging problem. On the one hand, as we all know that the industrial CPSs are referred as a vast and complex system, there are numerous industrial data streams created by various smart sensors and physical devices. On the other hand, another significant difficulty is that the probability of such anomalous events occurring in the real world is extremely low, thus leading to a scarcity of well-labeled network traffic for model training. Not only that, the reason for the lack of monitoring data may be resulted by a variety of factors like sensor errors, information transmission errors and so on (Humayed, Lin, Li, & Luo, 2017). These problems can make data acquisition and model training more difficult to achieve excellent anomaly detection. In addition, most of the nodes in industrial CPS are deployed in the resource-constrained devices, for example, limited electricity, low computational resource, bad communication environment as well as restricted storage capacity. Since industrial CPSs enable interaction with physical process through man-machine interfaces, the use of networked spaces to manipulate the physical entities in a reliable, low latency and safe way, therefore, high accuracy and low latency intrusion detection are often required to mitigate the damage caused by cyber hacking attacks in industrial CPS.

Therefore, it is worthwhile studying how to compress the size of the intrusion detection model as well as improve the generalization ability of the model without sacrificing its efficiency. Motivated by (Ok, Abe,

Miyao, & Kurita, 2020; Zhao, Chen, Wang, Shi, & Xue, 2021), we can use knowledge distillation (Hinton & Vinyals, 2015) and deep separable convolution to reduce model complexity. In addition, deep metric learning (Hoffer, 2015) is also introduced to enhance the performance of our intrusion detection model. We propose a lightweight yet efficient intrusion detection approach based on knowledge distillation (KD) and triplet convolution neural network (TCNN) named KD-TCNN for industrial CPS intrusion detection. In this paper, the complexity and computation of the proposed intrusion detection model can be reduced by three aspects to achieve higher accuracy and real-time anomaly detection: (1) For feature selection, the optimal feature subset of the dataset is selected by the binary grey wolf optimization algorithm. (2) For model compression, KD is used to transfer the inner-class information learned from the large model to the tiny model to improve the performance of the tiny model. (3) For convolution neural network architecture, the number of parameters and computational effort are reduced by using deep separable convolution neural network instead of traditional convolution neural network. This paper also proposes a new K-fold cross training method that uses the idea of pre-training and fine-tuning to further enhance the performance of the proposed KD-TCNN. Compared to traditional deep learning approaches and several state-of-the-art models, the KD-TCNN model has significant advantages in all performance metrics on the industrial CPS benchmark dataset NSL-KDD and CIC IDS2017. Furthermore, the complexity as well as the number of parameters of our model are significantly lower compared to the original model while the accuracy of intrusion detection on NSL-KDD dataset just decreases by 0.4%.

In general, the contributions of this study to the intrusion detection area are as follows:

- 1) A lightweight yet powerful model for industrial CPS intrusion detection based on knowledge distillation and deep metric learning is proposed. Based on our knowledge, we believe that there are few papers that use knowledge distillation for intrusion detection.
- 2) Deep separable convolution neural network which can both extract data features and reduce the computational burden at the same time is used in the student network. The advantage of deep metric learning is also applied for knowledge distillation to lower the gap between the teacher and student network outputs.
- 3) A robust model loss function is designed, which consists of the cross-entropy loss of student network outputs and labels, the KL divergence loss of teacher-student network, as well as the triplet loss of anchor-positive and anchor-negative.
- 4) A new neural network training method called K-fold cross training is proposed to enhance the anomaly detection capability of the model using the idea of pre-training and fine-tuning.

The remainder of the paper is organized as follows. Related techniques for industrial CPS intrusion detection and several intrusion detection models based on deep metric learning are discussed in Section 2. The framework for knowledge distillation models based on triplet convolution neural network for anomaly detection is presented in Section 3. The dataset utilized in this work, as well as its pre-processing, are described in Section 4. Section 5 introduces the experimental simulation environment, hyperparameter settings, baseline model, and model evaluation metrics. In Section 6, we prove the predominant performance of the proposed KD-TCNN through extensive experiments on the industrial CPS benchmark dataset NSL-KDD and CIC IDS2017. Finally, we present our conclusions and provide some future works for further investigation in the last section.

2. Related works

In this section, we will focus on the intrusion detection models applied in industrial CPS and introduce some anomaly detection models in recent years based on deep metric learning.

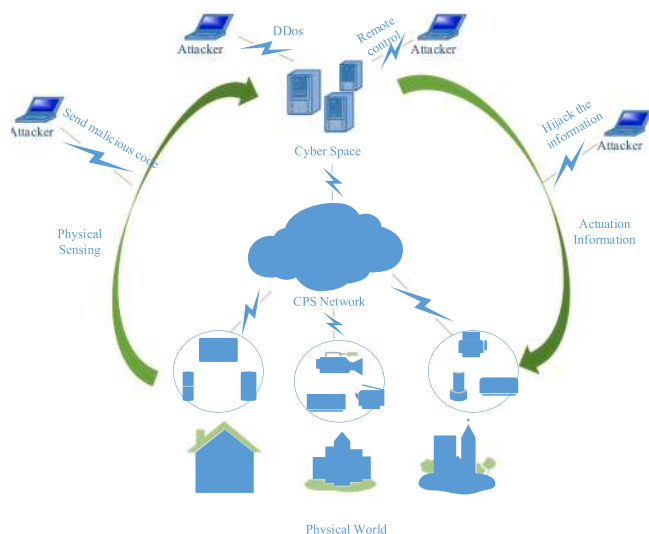


Fig. 1. General architecture of industrial CPS.

2.1. Intrusion detection solutions for industrial cyber-physical systems

In recent years, researchers not only use data mining algorithms, but also apply machine learning approaches as well as deep learning methods to detect abnormal network traffic in industrial CPS as the technology and science advanced (Olowononi, Rawat, & Liu, 2020; Trunzer, Kirchen, Folmer, Koltun, & Vogel-Heuser, 2017; Wickramasinghe, Marino, Amarasinghe, & Manic, 2018); and these methods have achieved good intrusion detection results in industrial CPS. However, in terms of data mining algorithms, a huge quantity of network traffic is needed to extract important features for intrusion detection, what's more, the intrusion detection results are extremely low for those attacks involving a small number of samples. Additionally, most data mining algorithms often have difficulty handling with the data set consisting of noisy data. Assuming that the data set contains lots of noisy data, it will tremendously affect the accuracy of the algorithm (Wang, Li, Wang, & Li, 2021). Presently, most researchers are willing to use a variety of machine learning-based approaches such as Logistic Regression (LR), Random Forest (RF), Naïve Bayes (NB), Gradient Boosting Decision Tree (GBDT), Artificial Neural Network (ANN), Ensemble Learning (EL) to identify anomaly network traffic in industrial CPS. Nonetheless, conventional machine learning approaches are regarded as shallow learning, these approaches also rely heavily on feature engineering and emphasize the importance of feature selection. As a result, they are unable to fully address large-scale intrusion detection classification problems that arise in real-world web application contexts. As the data set develops dynamically, the accuracy of the model dealing with multi-classification issues diminish drastically. As a consequence, a large number of researchers have tried to use deep learning approaches to address the vulnerabilities and security issues of industrial CPS. Li et al (Li, Wu, Song, Lu, Li, & Zhao, 2020) proposed an anomaly detection model for industrial CPS based on federated learning, which can construct a sophisticated and privacy-preserving anomaly detection model. Besides, a secure protocol which is able to maintain the model parameters' security and privacy during the training procedure was designed for federated learning, and experimental results showed that the model was good at classifying most industrial abnormal traffic. In order to select the optimal subset of features, Maha et al (Althobaiti, 2021) used the feature selection technique based on bbFO (binary bacterial foraging optimization) to reach the goal. They detected the intrusions in industrial CPS environments by utilizing the advantages of gated recurrent unit model, and the NADAM optimizer was used to improve the detection rate and tune the hyperparameters of the GRU. Experimental results indicated that the GRU model not only reduced running time, but also achieved an accuracy of 98.45% on the CICIDS-2017 dataset. Paulo et al (de Araujo-Filho, Kaddoum, Campelo, Santos, Macêdo, & Zanchettin, 2020) proposed a novel fog-based, unsupervised intrusion detection model for industrial CPS that reduced computational resources of the model as much as possible and so helps satisfy the demand for low latency requirements. The architecture introduced Generative Adversarial Networks (GAN), which computes the reconstruction error based on reconstructing network traffic samples transformed into the feature space to achieve a greater detection rate. Experimental results showed that the model not only achieved higher detection rate on some intrusion detection benchmark datasets, but also it was 5.5 times faster than the baseline method, as well as achieving the low latency requirements. Wang et al (Wang et al., 2018) proposed a two-stage intrusion detection model for smart grid with comprehensive and imperfect network traffic information. A security system based on interval state estimate is designed to successfully identify established network threats. Finally, nonlinear and nonstationary features were correctly extracted using a SAE (stacked autoencoder) model, which were then used to enhance the performance of the model.

Apparently, previous studies have shown that researchers have successfully used deep learning techniques to identify anomalous network traffic in industrial CPS. However, it is well known that well

labelled training samples and prior knowledge play a significant role in traditional deep learning models. Not only that, the number of the deep learning model parametric number determines how good the model is. Limited by the constrained power, computation, communication and storage capacity of the nodes in the IoT network, it is difficult to deploy large models to each node in the IoT network.

2.2. Deep metric learning based intrusion detection model

Through training and learning, the goal of deep metric learning is to minimize or restrict the distance between similar samples while increasing the distance between different classes of data (KAYA, Mahmut, and Hasan Ş. BİLGE, , 2019). Usually, Siamese Neural Network (Bromley, Guyon, Lecun, Säckinger, & Shah, 1993) and Triplet Neural Network (Hoffer, 2015) are two commonly used neural network architectures for metric learning, which are also widely applied in the domain of intrusion detection. For example, so as to analyse the properties of flow-based network traffic data, Andresini et al (Andresini, Appice, & Malerba, 2021) proposed a new intrusion detection approach based on deep metric learning. The proposed model combined autoencoder with triplet neural network to improve the performance of intrusion detection. Two independent autoencoders were trained by utilizing normal and abnormal traffic in the training phase. After that, the triplet neural network was trained to get the feature representations of the network traffic. The advantage of these feature representations was that they could force the input network traffic to be recovered close to its reconstruction of the same category as well as distant from its reconstruction of the different category through the autoencoder. Network traffic in test set was allocated to the category affiliated with the flow which was the nearest reconstruction of the new flow in the latent space restored by the autoencoder during the prediction stage. As a result, the prediction phase was able to achieve good prediction performance in identifying malicious network traffic by using the embeddings gained during the training phase. Bedi et al (Bedi, Gupta, & Jindal, 2021) proposed an ensemble method to solve the class imbalance issue of intrusion detection dataset from the algorithm level rather than use the data-level balancing techniques. To identify abnormal traffic, the first layer used an ensemble of binary extreme Gradient Boosting, Siamese Neural Network as well as deep neural network to classify the input data into normal and abnormal network traffic, and subsequently fed the abnormal traffic identified from the previous layer into the next layer for classifying into different abnormal categories through the multi-class extreme Gradient Boosting classifier. The experimental results indicated that the proposed method outperformed traditional deep learning algorithms and the state-of-the-art models in terms of recall, precision and other performance metrics on several intrusion detection datasets without using other oversampling methods. To address the data imbalance problem in the intrusion detection datasets as well as improve the detection rate of zero-day attacks, Wang et al (Wang, Tian, Qin, Fang, & Chen, 2021) proposed an anomaly detection model utilizing the Siamese Capsule Neural Network based on few-shot learning. Furthermore, an unsupervised subtype sampling technique was used to address the data imbalance problem that existed in most intrusion detection datasets, and the Siamese Capsule Neural Network excelled at learning good feature representations. The experimental results demonstrated that only a very small number of network traffic samples used to train the model could obtain a good classification performance in detecting both known and unknown attacks, which was particularly prominent in the detection of unknown attacks thanks to the advantages of the deep metric learning.

In conclusion, machine learning methods and deep learning approaches both made significant advances in the domain of intrusion detection as time and technology progressed. In contrast to the previous intrusion detection methods for industrial CPS, we propose an intrusion detection model based on knowledge distillation and triplet convolution neural network, which is applied to the industrial CPS benchmark dataset NSL-KDD and CIC IDS2017, taking into full consideration the

limited power, low computational resource, bad communication environment and restricted storage capacity of the nodes in IoT networks as well as the model generalization capability. A variety of evaluation metrics such as accuracy, precision, detection rate, FMeasure, model size and number of model parameters are applied to validate the performance of the KD-TCNN model, which makes the assessment of the proposed KD-TCNN more logical as well as far reaching.

3. Methodology

In this section, we focus on the framework of knowledge distillation intrusion detection model based on triplet convolution neural network, which can be divided into four parts: (1) Feature selection. (2) Knowledge distillation based on triplet convolution neural network. (3) Deep separable convolution neural network. (4) K-fold cross training. The detailed implementation of each of these four parts and how to fuse the knowledge distillation model based on triplet convolution neural network to intrusion detection, and the overall intrusion detection system process will be also discussed in this section.

3.1. Feature selection

As we all know, feature selection plays a significant role in feature engineering, and the purpose of feature selection is to determine the best subset of the features. Feature selection not only can help to minimize the number of features, improve the performance of the model, reduce running time of the model, but also it can avoid the overfitting problem by eliminating irrelevant or redundant features. According to the form of feature selection, there are three main different types of feature selection algorithms: (1) Embedded, firstly, the weight coefficients of each feature can be obtained by employing machine learning methods in training stage, which are then used to rank the features from small to large based on these coefficients obtained from the training stage. (2) Filter, scoring each feature by dispersion and relevance, setting a threshold or the number of features and then select features according to the threshold. (3) Wrapper, according to the objective function, several features are selected or excluded at a time. Since the evaluation criteria of the filtered feature selection algorithm are unaffected by the specific learning algorithm, the selected feature subset often has lower classification performance than the wrapper method, moreover, the embedded feature selection algorithm is usually more sensitive to anomalies and easily falls into the local optimal point, so the wrapper feature selection algorithm is considered in this paper to find the best feature subset. Because grey wolf optimization method (Mirjalili, Mirjalili, & Lewis, 2014) has the benefits of easy implementation, few adjustment parameters required, simple structure and excellent performance in accuracy as well as convergence speed of the model. While the solution is restricted to binary values $\{0, 1\}$ when the grey wolf optimization algorithm is applied to find the optimal feature subset, so we use the binary version of grey wolf optimization (Emary, Zawbaa, & Hassanien, 2016) (bGWO) to select the optimal feature subset. The grey wolf optimization algorithm is a bio-inspired optimization algorithm that simulates grey wolf predatory behaviour by assigning predatory tasks such as encirclement, pursuit, and attack to different levels of grey wolf packs in order to complete predatory behaviour according to grey wolf social rank, thus realizing the process of global optimization.

In order to describe the grey wolf optimization algorithm, the social hierarchy of the grey wolves was modelled as follows. We named the most powerful wolf as α , the subordinate wolf and the third level of the structure of the wolves are defined by β and δ , while the remaining wolves are named ω . The hunting process is guided by α, β and δ wolves, and ω wolves follow these three kinds of wolves in the grey wolf optimization method. The specific steps of the grey wolf optimization algorithm are shown below.

Step 1. Prey encirclement stage.

In order to hunt the prey, the grey wolves usually surround the prey firstly. Equations (1)-(4) are used to mathematically show the encircling behavior.

$$X[t+1] = X[t] + A \bullet D \quad (1)$$

$$D = C \bullet X_p[t] - X[t] \quad (2)$$

$$A = 2\alpha \bullet r_1 - \alpha \quad (3)$$

$$C = 2r_2 \quad (4)$$

where t is the iteration number, A is defined by Equation (3) and C is defined by Equation (4), X_p is the prey position, the grey wolf's current and next positions are defined by $X[t]$ and $X[t+1]$. α is applied for the trade-off between exploitation and exploration, the updating equation for α is defined by Equation (12), r_1 and r_2 are random vectors that initialize with values in $[0, 1]$.

Step 2. Hunting stage.

The α wolves are referred as leader in hunting stage, while the β and δ wolves might also take part in decision-making process in hunting stage occasionally. Other search agents (including the ω) are forced to change their locations in accordance with the best search agents' location obtained so far. And Equations (5)-(8) define the updating equations for the wolves' locations.

$$X[t+1] = (X_1[t] + X_2[t] + X_3[t])/3 \quad (5)$$

$$X_1 = X_\alpha - A_1 \bullet D_\alpha \quad (6)$$

$$X_2 = X_\beta - A_2 \bullet D_\beta \quad (7)$$

$$X_3 = X_\delta - A_3 \bullet D_\delta \quad (8)$$

where X_α, X_β and X_δ represent the location of α, β and δ at a given iteration t . A_1, A_2 and A_3 are obtained using Equation (3). D_α, D_β and D_δ are calculated using Equations (9)-(11), respectively.

$$D_\alpha = C_1 X_\alpha - X \quad (9)$$

$$D_\beta = C_2 X_\beta - X \quad (10)$$

$$D_\delta = C_3 X_\delta - X \quad (11)$$

where C_1, C_2 and C_3 are defined by Equation (4) mentioned above.

Step 3. Attacking stage.

The grey wolves will end the hunting stage by raiding the prey when it stops moving. In order to simulate the process of approaching prey, α mentioned above is updated in each round according to Equation (12), decreasing from 2 to 0.

$$\alpha = 2 - t(2/MaxIteration) \quad (12)$$

where t is the current iteration number and the maximum number of the iterations is $MaxIteration$.

While in binary grey wolf optimization algorithm, the wolves updating equations are based on three location vectors called x_α, x_β and x_δ , which draws each wolf towards the initial three optimal solutions. In our paper, the second version of the binary grey wolf optimization algorithm will be adopted, which is named bGWO2 (Emary and Zawbaa, 2016). The grey wolf position updating formula is shown in Equation (13).

$$x_d^{t+1} = \begin{cases} \text{if sigmoid}\left(\frac{x_1 + x_2 + x_3}{3}\right) \geq \text{rand} \\ 0 \text{ otherwise} \end{cases} \quad (13)$$

where *rand* is a random number sampled from the uniform distribution of [0,1], the updating binary position in dimension *d* at the iteration *t* is denoted by x_d^{t+1} , Equation (14) shows the calculation formula of *sigmoid*(***).

$$\text{sigmoid}(x) = 1/(1 + e^{-10(x-0.5)}) \quad (14)$$

The bGWO algorithm adaptively searches the feature space to find the optimal feature subset with the best classification performance and the lowest number of features picked. Equation (15) shows the fitness function applied to evaluate the position of each grey wolf in bGWO.

$$\text{Fitness} = \alpha * (1 - P) + \beta * \frac{L}{T} \quad (15)$$

where *P* is the performance metric of the model, *L* is the number of elements in the optimal feature subset, while *T* is the total number of elements in the feature set. The significance of classification accuracy and the optimal feature subset length are represented by the parameters *alpha* and *beta* and $\alpha \in [0, 1]$, $\beta = 1 - \alpha$.

We use the training set to apply the feature selection of the network traffic. The learning method in the feature subset evaluation step is convolution neural network, and Equation (15) is used as the fitness function. In order to achieve feature selection and dimensionality reduction, the feature subset with the lowest fitness function value is chosen, resulting in the feature subset with the optimal intrusion detection performance. Fig. 2 shows the flow chart of the entire binary grey wolf optimization feature selection algorithm.

3.2. Knowledge distillation based on triplet convolution neural networks

Knowledge distillation is a common approach to model compression, which refers to distilling the feature representations learned by a complex, highly capable teacher network, and passing them to a small, weakly capable student network. Then we can get a student network with fast speed, strong ability but few parameters. From another perspective, the outputs of the student model are brought as near to the teacher model as feasible through knowledge distillation, which can enable the student network to learn more softened knowledge from the teacher network, which contains inter-category information that is not available in traditional one-hot encoding. Because knowledge distillation tries to minimize the difference between the teacher network and the student network, while deep metric learning tries to reduce the difference between related inputs and increase the difference between different input samples at the same time. Therefore, knowledge distillation can utilize the ability of deep metric learning to minimize the difference between comparable inputs to lower the gap between the teacher and student network outputs, hence boosting the student model's performance. Generally, Siamese Neural Network and Triplet Neural Network are two common neural network architectures for metric learning. Because the Siamese Neural Network may only assess the distance between two samples, it must decide the notion of similarity

between two samples in a unique way. For example, based on the notion of gender, two distinct guys should be evaluated as similar. They should, nevertheless, be assessed as dissimilar based on the notion of individuality. It's too difficult for the Siamese Neural Network to express these multiple concepts, however, Triplet Neural Network learns to reduce the gap between anchor-positive and anchor-negative, allowing multiple related concepts to be evaluated without relying on a single notion, so the Triplet Convolution Neural Network is chosen to reduce the distance between the teacher and the student model outputs in our paper. Our proposed KD-TCNN aims to address the problem of limited computing power and resources of IoT nodes in industrial CPS, and Fig. 3 illustrates the knowledge distillation intrusion detection model based on triplet convolution neural network.

It can be seen in Fig. 3, so as to train the KD-TCNN model, the sampled data x_a from the training set is fed into the pre-trained teacher model, and the output $z_{a_i}^t$ from the softmax layer is then applied for calculating the probability vector $q_{a_i}^t$ for each category, which can be described as follows:

$$q_{a_i}^t = \frac{\exp(z_{a_i}^t/T)}{\sum_j \exp(z_{a_j}^t/T)} \quad (16)$$

where *T* is the temperature which is generally set to 1. When *T* is set to a larger value, the probability distribution over classes becomes softer.

The sampled data x_a and the x_n which belongs to different class of x_a are fed into the student network, and the softmax layer outputs $z_{a_i}^s$ and $z_{n_i}^s$ to calculate the probability vectors $q_{a_i}^s$ and $q_{n_i}^s$ for each category, which can be expressed as follows:

$$q_{n_i}^s = \frac{\exp(z_{n_i}^s/T)}{\sum_j \exp(z_{n_j}^s/T)} \quad (17)$$

$$q_{a_i}^s = \frac{\exp(z_{a_i}^s/T)}{\sum_j \exp(z_{a_j}^s/T)} \quad (18)$$

To ensure the prediction accuracy of anomaly detection for industrial CPS data, three losses are considered in the design of our loss function, which consisted of a triplet loss L_{triplet} based on anchor-positive and anchor-negative, a cross-entropy loss L_{hard} between the student network outputs and the ground truth, and a KL divergence loss L_{soft} of the teacher-student network.

For the same sample, the anchor samples and positive samples come from the outputs of the teacher network and the student network, respectively. Similarly, the negative samples are those samples whose outputs from the student network differs from the positive sample class. It's universally acknowledged that the advantages of the triplet loss is that it is useful for reducing the difference between the anchor-positive outputs as well as increasing the difference between the anchor-negative outputs. We introduce this technique into knowledge distillation and the triplet loss can be obtained from above formula as follows:

$$L_{\text{triplet}} = \sum_{(a,n) \in \Omega} \max(0, m + \|q_{a_i}^t - q_{a_i}^s\|_2^2 - \|q_{a_i}^t - q_{n_i}^s\|_2^2) \quad (19)$$

where *m* is margin, a hyperparameter, Ω is the intrusion detection data

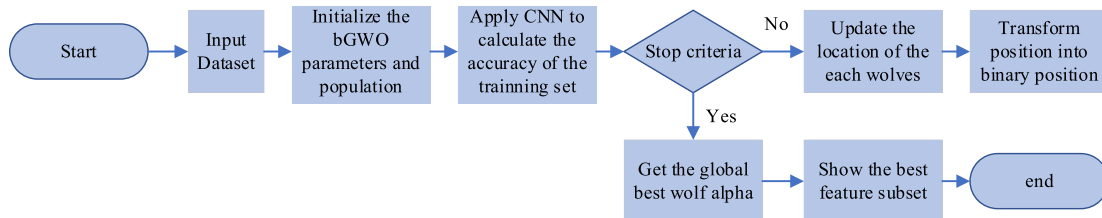


Fig. 2. The flow chart of binary grey wolf algorithm.

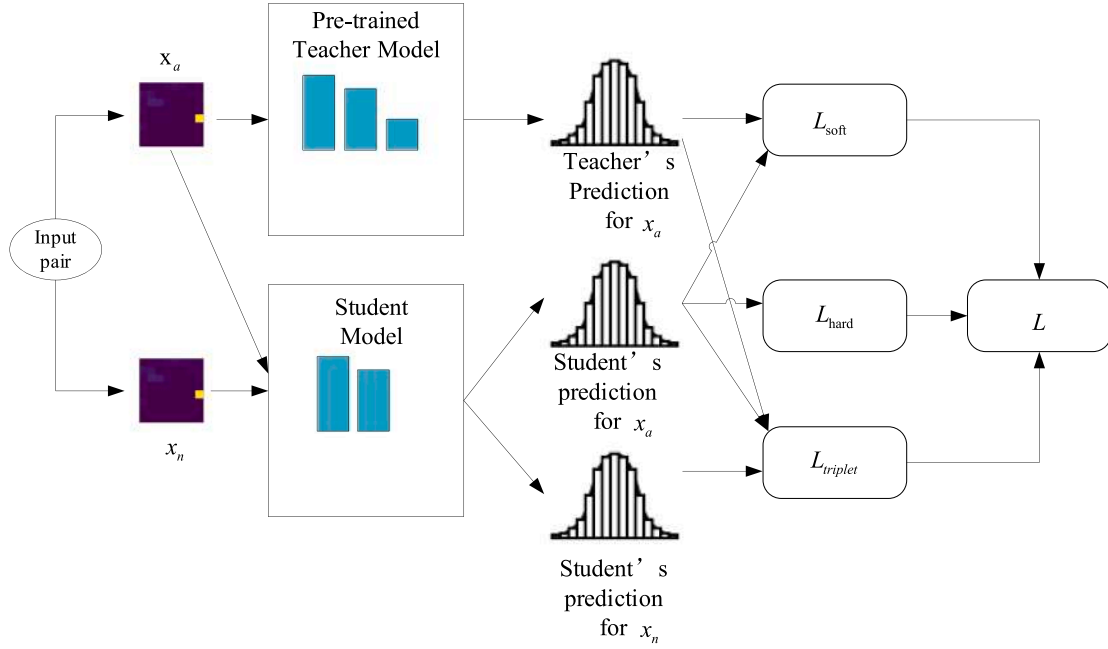


Fig. 3. The knowledge distillation framework based on triplet convolution neural network.

sets of industrial CPS.

In order to lower the gap between the teacher and student network outputs, the KL divergence of teacher networks' softmax outputs and student network's softmax outputs will be used as a part of the model loss function, and the KL divergence loss L_{soft} of teacher-student network can be calculated from the above formula as follows:

$$L_{soft} = \sum_{(a,n) \in \Omega} KL(q_{a_i}^t, q_{a_i}^s) \quad (20)$$

where $KL(p, q)$ is the KL divergence loss between the student network's outputs and the teacher network's outputs, and the formula for calculating KL divergence is defined as follows:

$$KL(p, q) = \sum_i p_i \log \left(\frac{p_i}{q_i} \right) \quad (21)$$

To constrain the degree of difference between the student network's outputs and the probability distribution of the ground truth, we use the cross-entropy loss of the student network's outputs and the ground truth as a part of the model loss function, and define the cross-entropy loss L_{hard} of the student network's outputs and the ground truth as follows:

$$L_{hard} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log p_{i,k} \quad (22)$$

where $y_{i,k}$ means that the i -th sample's label is k , $p_{i,k}$ denotes the probability that the i -th sample is predicted to be label k , N is the total number of network traffic and K is the total number of categories.

Inspired by the paper (Li, 2022), we define the loss of knowledge distillation L_{KD} as follows:

$$L_{KD} = \alpha T^2 * L_{hard} + (1 - \alpha) * L_{soft} \quad (23)$$

where T is the temperature used to soften the probability distribution discussed above, and α is the hyperparameter used for balancing between the two components of the KD loss.

Since the model loss function consists of multiple components, we need to add coefficients to the loss terms to adjust the contribution of each loss to the overall loss function. As a result, the model loss function L can be calculated using the following formula:

$$L = L_{KD} + \theta L_{triplet} \quad (24)$$

where θ is the balance coefficient that controls the knowledge distillation loss and the triplet loss during the model training stage. We adjust the training error according to Equation (24) until the student model reaches convergence, and save the optimal student model for later test experiments.

3.3. Deep separable convolution neural network

The main notion of deep separable convolution is to divide a whole conventional convolution process into two phases, which go through a depthwise convolution and a pointwise convolution respectively, just as explained in Xception (Chollet, 2017) and MobileNet (Howard, Zhu, Chen, Kalenichenko, Wang, Weyand, Andreetto, & Adam, 2022). Fig. 4 (a) and (b) show the conventional convolution operation and the deep separable convolution operation, respectively.

One channel is accountable for one convolution kernel of the depthwise convolution, and one channel is convolved by just one convolution kernel, hence the output's feature map processed by depthwise convolution precisely has the same number of channels as the input layer, so the number of parameters of the depthwise convolution can be calculated from the above formula as follows:

$$num_params = W_k * H_k * in_channels \quad (25)$$

And the computational complexity can be obtained by following formula:

$$Flops = W_k * H_k * (W_{img} - W_k + 1) * (H_{img} - H_k + 1) * in_channels \quad (26)$$

where W_k and W_{img} are the width of the kernel and the image, H_k and H_{img} are the height of the kernel and the image, $in_channels$ means the number of input channels.

After being processed by depthwise convolution, the output has the same number of feature maps as the input layer, which cannot expand feature map size. Moreover, because the depthwise convolution performs the convolution operation separately for each channel of the input layer, it is ineffective in combining feature information from different channels at the same spatial location, so we can combine these feature

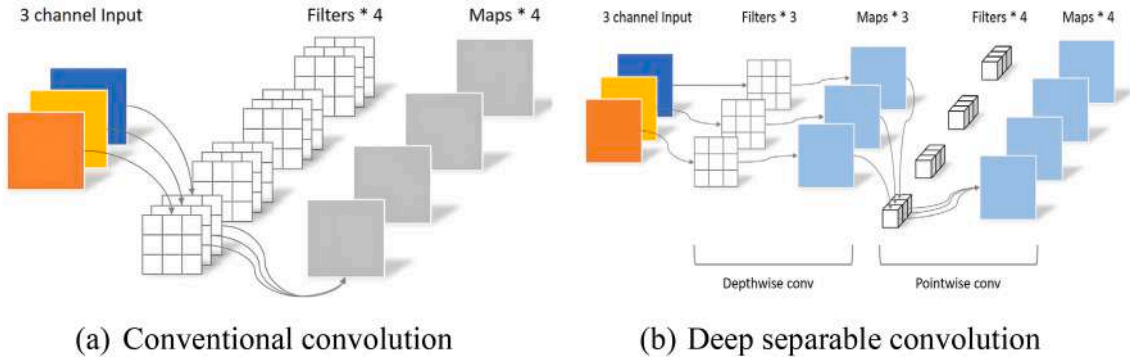


Fig. 4. Comparison of conventional and deep separable convolution.

maps to generate a new feature map through the pointwise convolution.

The pointwise convolution works in a similar way to the regular convolution operation, while pointwise convolution has a convolution kernel of size $1 \times 1 \times M$, and M is the number of channels in the previous layer. In order to generate a new feature map, the pointwise convolution here will merge the feature maps from the previous phase in the depth direction, so the number of parameters of pointwise convolution can be expressed as follows:

$$\text{num_params} = 1 \times 1 \times \text{in_channels} \times \text{out_channels} \quad (27)$$

And the computational complexity can be obtained by following formula:

$$\text{Flops} = 1 \times 1 \times W_{\text{feature}} \times H_{\text{feature}} \times \text{in_channels} \times \text{out_channels} \quad (28)$$

where W_{feature} and H_{feature} are the width and height of the feature map, out_channels denotes the number of output channels.

By decomposing the regular convolution operation into two steps, the number of parameters as well as the computational resources of the deep separable convolution greatly decreased. For example, suppose that the input feature map size is $256 \times 256 \times 16$, the output feature map size is $256 \times 256 \times 32$, the convolution kernel size is 3×3 , if the conventional convolution is used, then the number of parameters used is $3 \times 3 \times 16 \times 32 = 4608$, and the computational complexity is $3 \times 3 \times (256 - 2) \times (256 - 2) \times 16 \times 32 \approx 297$ million. If the deep separable convolution is introduced in convolution neural network, then the number of parameters used is $3 \times 3 \times 16 + 1 \times 1 \times 16 \times 32 = 656$, and the computational complexity is $3 \times 3 \times (256 - 2) \times (256 - 2) \times 16 + 3 \times 3 \times 16 \times 32 \approx 9.3$ million. Obviously, the computation and number of parameters of deep separable convolution are much smaller than those of conventional convolution, so deep separable convolution can be applied to intrusion detection models, which then can be deployed to nodes with limited computational power in IoT networks, thus significantly reducing the intrusion detection time.

3.4. K-fold Cross training

In our paper, we propose a new neural network training method called K-fold cross training, which is a little similar to K-fold cross validation. K-fold cross training divides the training set into K parts equally, then each subset is made into a validation set once separately, and the rest $K - 1$ subsets are referred as the training set. Unlike K-fold cross validation, each K-fold cross validation yields K models, and the performance metric for the classifier is the average of the classification accuracy of the test set of these K models, while K-fold cross training only yields one model, and the model is continuously tuned on the basis of the previous training, similar to the idea of pre-training and fine-tuning, so that the model has strong prior knowledge instead of random initialization before training and continues to being fine-tuned on the basis of the previous step until the end of the training cycle, which

can allow the model to converge faster as well as avoid falling into a local optimum. As the learning rate decreases, the updating speed of the model slows down, which makes it easier for the model to reach the global optimum. In this way, although the training time of the model will be longer, the accuracy of the model can be significantly improved just as shown in Table 13. The specific steps of K-fold cross training are as follows: (1) Define the model and learning rate, and divide the dataset into training and test datasets. (2) Divide the training set into K parts, one part is used as the validation set, and the remaining $K - 1$ parts are used as the training set. (3) Define a gradient optimizer with a decay strategy for learning rate, model training is done with $K - 1$ subsets, and model testing is done with the rest subset. (4) Repeat step (3) K times to obtain the optimal model as well as get the accuracy of the model using the validation dataset. The pseudo code for K-fold cross training algorithm steps is shown in Algorithm 1.

Algorithm 1 Pseudocode of K-fold cross training in a PyTorch-like style.

```
# student, teacher: student and teacher model in KD-TCNN
# (X_train, Y_train), (X_test, Y_test): Training set and Testing set
# LEARNING_RATE: Learning rate for optimizer
# EPOCHS: Total epochs for training
# FLAG: Learning rate decay
for x_train, y_train, x_val, y_val in splits(X_train, Y_train): //Divide the training set into
    K folds
        optimizer = Adam(LEARNING_RATE / FLAG)
        train_loader = DataLoader(x_train, y_train)
        val_loader = DataLoader(x_val, y_val)
        for epoch in range(EPOCHS):
            Train(student, teacher, train_loader, optimizer) //Train KD-TCNN model
            Val(student, val_loader)
        FLAG = FLAG * 5 //Learning rate decay
    Results = Test(student, X_test, Y_test)
```

3.5. The intrusion detection steps of KD-TCNN model

In this paper, we use KD-TCNN model for industrial CPS intrusion detection, and the datasets applied for anomaly detection are NSL-KDD and CIC IDS2017. Through the discussions in Sections 3.1–3.4, the detailed steps of the KD-TCNN model for intrusion detection are as follows:

- Step 1. Applying numerical processing of character-based data, data normalization and data imbalance processing for intrusion detection datasets, please refer to Section 4.2 for further information.
- Step 2. Selecting the optimal feature subset of the intrusion detection dataset using the binary grey wolf optimization algorithm.
- Step 3. Pre-training the teacher model based on the optimal feature subset.
- Step 4. Intrusion detection model KD-TCNN training:
 - Set up the model parameters and figure out how the student network is structured.

Two different classes of network traffic based on the optimal feature subset are fed into the KD-TCNN for model training.

The error of the K-fold cross training process is modified according to the cost function derived from Equation (24) until the student network model reaches convergence.

Step 5. Testing for KD-TCNN model, the pre-processed test dataset is fed into the student network to obtain the classification results for each network traffic.

The overall flow chart of the KD-TCNN model is shown in Fig. 5 below.

4. Datasets and pre-processing

For the purpose of examining the detection capability of the KD-TCNN model for industrial CPS anomaly detection, our paper conducts anomaly detection experiments on industrial CPS benchmark datasets NSL-KDD (L.Dhanabal, Dr. S.P. Shantharajah. "A Study on NSL-KDD Dataset for Intrusion Detection System Based on Classification Algorithms," International Journal of Advanced Research in Computer and Communication Engineering. Vol. 4, Issue 6, June 2015, 2015) and CIC IDS2017 (Iman Sharafaldin, Arash Habibi Lashkari, Ali A. Ghorbani. "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization," 4th International Conference on Information Systems Security and Privacy (ICISSP), Portugal, January 2018, 2018).

4.1. Dataset description

The NSL-KDD dataset addresses concerns of duplicate data and data redundancy that plagued the KDD CUP99 dataset. Four sub datasets are contained in the NSL-KDD dataset, named KDDTrain+.txt, KDDTest+.txt, KDDTrain-21.txt and KDDTest-21.txt. In order to train KD-TCNN

model better, we choose KDDTrain+.txt and KDDTest+.txt as our experimental training set and test set. Compared with anomaly traffic, the intrusion detection dataset of emerging CPS contains a large number of benign samples, so we can use the NSL-KDD dataset to analogize the network traffic in the real CPS environment and evaluate our model by analyzing different anomaly traffic detection. Table 1 describes the detailed description of the NSL-KDD dataset.

Communications Security Establishment and the Canadian Institute for Cybersecurity evaluated 11 datasets available since 1998 and found that most of them are outdated and unreliable. Traffic diversity is neglected by some famous intrusion detection datasets, some of them lack a wide range of known attacks, and others do not utilize packet payload data effectively. However, the CIC IDS2017 dataset provides network traffic data which includes benign and up-to-date typical Dos attacks that are similar to real-world traffic. Due to the diversity of industrial CPS devices, they are vulnerable to a variety of network attacks, including DDoS and SQL injection attacks, etc. The CIC IDS2017 dataset contains a large number of Dos attacks, so the CIC IDS2017 can be used as a benchmark dataset for industrial CPS intrusion detection. The Wednesday-workingHours.csv is used as the industrial CPS intrusion detection dataset in this paper. Since the number of Heartbleed attacks

Table 1
Network traffic distribution of NSL-KDD.

Attack category	Train	Test
Normal	67,343	9711
Probe	11,656	2421
Dos	45,927	7460
U2R	52	67
R2L	995	2885
Total	125,973	22,544

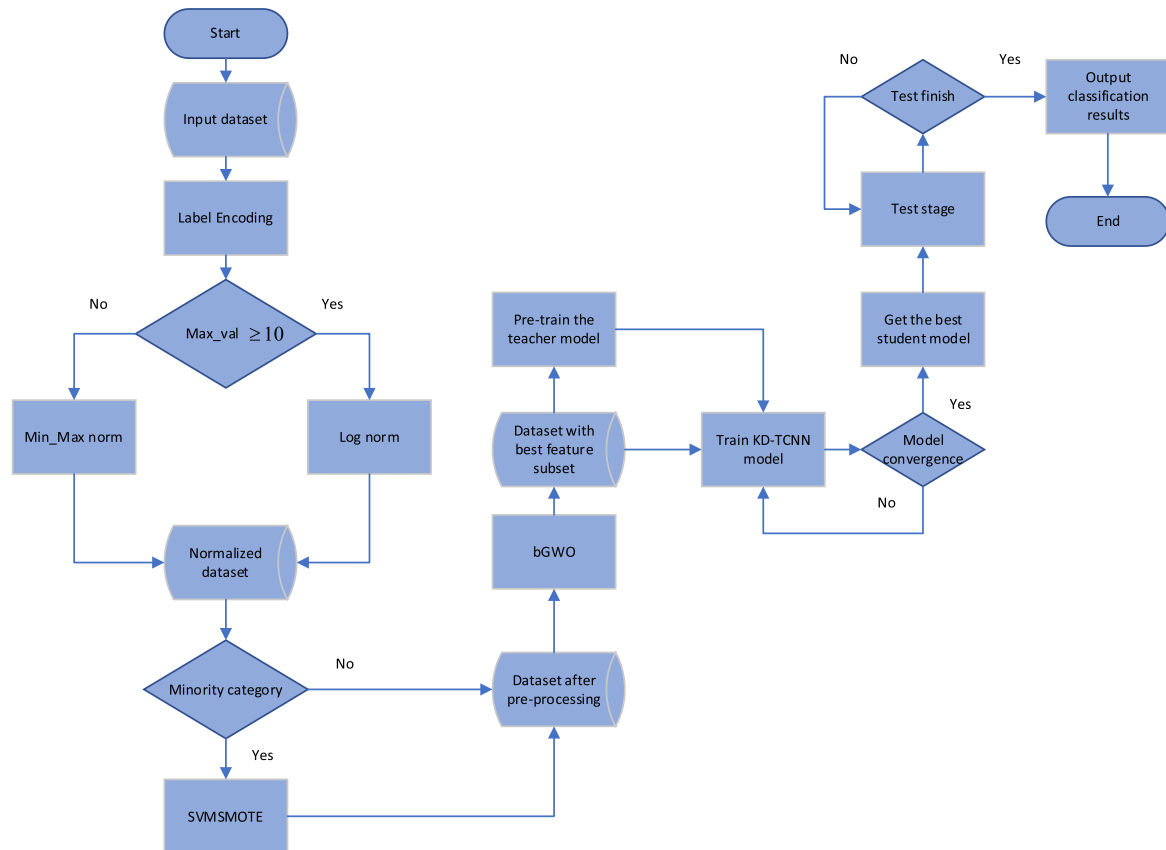


Fig. 5. Flow chart of the KD-TCNN model.

in Wednesday-workingHours.csv is only 11, which is too sparse, then this category is eliminated and the training and testing datasets are repartitioned. The detailed description of the CIC IDS2017 dataset is shown in Table 2.

4.2. Data pre-processing

The experimental datasets must be pre-processed since the original dataset must comply to the convolution neural network's input format and it's an important step for intrusion detection. The pre-processing stages are as follows:

Numerical processing of character data

Let's take NSL-KDD dataset as an example, there are three symbolic features in NSL-KDD dataset, which are protocol_type, flag and service, and we should convert these symbolic features into numeric features. For instance, there are three kinds of attributes in the protocol_type feature, which are TCP, UDP and ICMP, then the three categories can be processed as 0, 1, 2. We also process other symbolic features similarly as the protocol_type does. Then we need to perform reshape operation for network traffic and convert the network traffic of the NSL-KDD dataset to 8×8 grayscale graph format and the CIC IDS2017 dataset to 10×10 grayscale graph format.

Data normalization

Because the range of values per feature in the intrusion detection datasets varies widely, the dataset needs to be normalized. Min-Max normalization is a linear scaling method for pre-processing data that is commonly used in machine learning. However, Min-Max normalization depends on the sample's minimum and maximum values, so it has apparent drawbacks (Tang, Tan, & Chen, 2018). Inspired by the paper (Jin, Chen, Zhang, Yuan, & Wang, 2021), to address the circumstance when the value of per feature varies dramatically, we need a new scaling mechanism. Since the values of each feature in the NSL-KDD and CIC IDS2017 datasets vary dramatically, we determine to use a hybrid data pre-processing method and our data normalization method is shown in Equation (29).

$$\bar{x}_i = \begin{cases} \frac{x_i - x_{i_{min}}}{x_{i_{max}} - x_{i_{min}}}, & x_{i_{max}} < 10 \\ \log_{10}(1 + x_i), & x_{i_{max}} \geq 10 \text{ and } x_{i_{min}} > -1 \end{cases} \quad (29)$$

where $\bar{x}_i$ denotes the value of i -th feature after data normalization, x_i represents original value of the i -th feature, $x_{i_{max}}$ is the max value among the same feature, and $x_{i_{min}}$ represents the min value among the same feature.

Data imbalance

As can be seen from Tables 1 and 2, some malicious attack account for just a small fraction of all samples in industrial CPS intrusion detection scenarios. For example, there are serious data imbalance problems in the NSL-KDD dataset, where R2L and U2R attacks only account for 0.79% and 0.041% of the training set respectively, so the

Table 2
Network traffic distribution of CIC IDS2017.

Attack category	Train	Test
Benign	351,719	87,964
GoldenEye	8307	1986
Hulk	183,998	46,126
Slowhttptest	4444	1055
Slowloris	4648	1148
Total	553,116	138,279

classifier tends to be biased towards those categories with more samples, resulting in a large false alarm rate. To alleviate this problem, we use the SVMSMOTE (Nguyen, Cooper, & Kamei, 2009) algorithm to oversample those attack categories with small number of samples, but we only use this technique for the training set without changing the data distribution in the test set (To avoid over-reliance of the model on the generated data).

5. Experimental setup

The experiments in this paper are simulated under Windows 10 operating system with AMD Ryzen 5 3600 processor, 16 GB RAM and an NVIDIA RTX 2070 GPU.

5.1. Evaluation metrics

Due to the complexity of network intrusion detection dataset as well as obvious data imbalance problem, as a result, the assessment metrics for intrusion detection models in this study will include accuracy, precision, detection rate (DR), and FMeasure. Through a complete comparison of the above measures in a bunch of experiments, we can check the model's efficiency and stability more comprehensive, what's more, it is also sufficient to prove the superiority of our proposed model. In addition, we utilize the model size and number of parameters as assessment metrics for the intrusion detection model to quantify the computing resources required by the model as well. The above metrics are calculated using Eqs. (30)–(33):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (30)$$

$$Precision = \frac{TP}{TP + FP} \quad (31)$$

$$DR = \frac{TP}{TP + FN} \quad (32)$$

$$FMeasure = \frac{2 * Precision * DR}{Precision + DR} \quad (33)$$

where TP, TN, FP and FN mentioned above denote true positives, true negatives, false positives and false negatives respectively.

5.2. Hyperparameter setting

The KD-TCNN model has a few hyperparameters that need to be set artificially, and the selection of their values will definitely have a significant impact on the model's classification performance. Therefore, we need to determine their optimal values through a bunch of experiments. The hyperparameters in this paper are the temperature T for softening the probability distribution that appears in Section 3.2, the margin m in the triplet loss, the weight coefficient α that constrains L_{hard} and L_{soft} , as well as the equilibrium coefficient θ that controls the knowledge distillation loss and the triplet loss. It is shown through several experiments that when T is taken as 6, α is taken as 0.8, m is taken as 0.1 and θ is taken as 0.5 are the optimal solution of hyperparameters. In the next experiments, we used Adam (Kingma & Ba, 1412) as the gradient optimizer with an initial learning rate of 0.0001; Swish (Ramachandran, Zoph, & Quoc, 1710) as the activation function; NSL-KDD dataset batch size is set to 512, CIC IDS2017 dataset batch size is set to 1024, epoch is set to 40. We conducted the experiments with 5-fold cross validation and 5-fold cross training. The teacher and student model structures are shown in Fig. 6 and Fig. 7.

5.3. Model baseline

Based on the above hyperparameters, we input the pre-processed

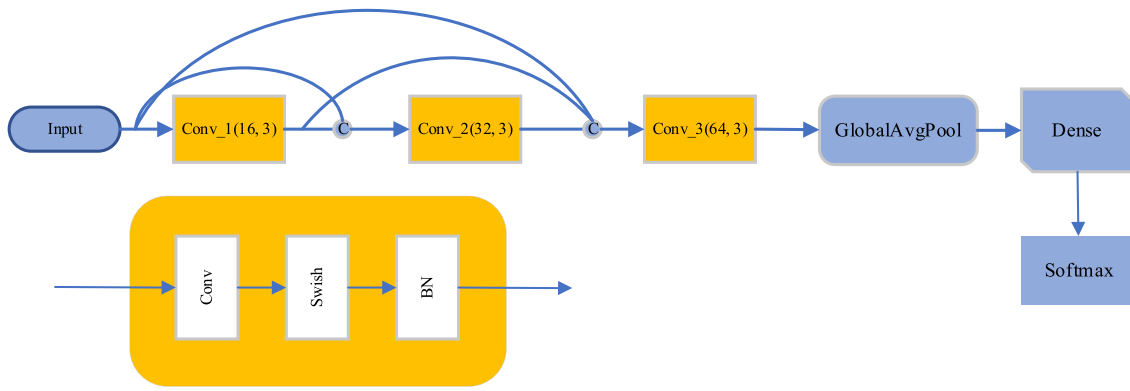


Fig. 6. Teacher model structure, 16/32/64 means the number of output channels, 3 denotes the size of kernels, BN means the batch normalization layer, C means concatenation.

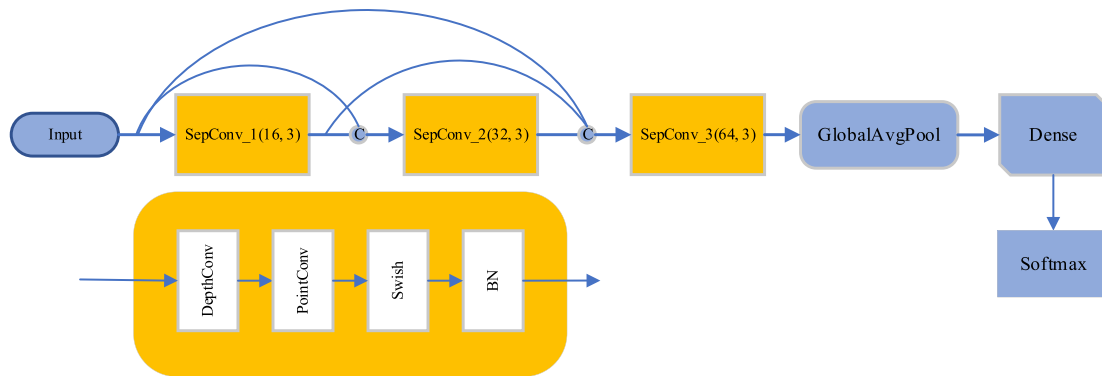


Fig. 7. Student model structure, 16/32/64 means the number of output channels, 3 denotes the size of kernels, BN means the batch normalization layer, C means concatenation.

NSL-KDD and CIC IDS2017 into teacher and student networks, then take the experimental results as baseline. We will compare the Baseline with the results of subsequent experiments later. The baseline of the NSL-KDD can be seen in Table 3, and the baseline of the CIC IDS2017 is shown in Table 4. As can be seen from Tables 3 and 4, since the teacher model has seven times more parameters than the student model, the teacher model exceeds the student model in all performance indicators. What's more, as the complexity of the data set increases and the number of network traffic decreases, the difference of performance between the teacher model and student model also increases.

6. Experimental results and discussion

We will conduct extensive experiments on the industrial CPS intrusion detection datasets NSL-KDD and CIC IDS2017 to show the effectiveness of our proposed KD-TCNN model, and multiple evaluation metrics described in Section 5.1 are used to measure the superiority of the KD-TCNN model with traditional deep learning approaches and the recent state-of-the-art models.

6.1. Comparison of feature selection algorithms

We validate the benefits of the bGWO and compare it with the Flower

Pollination Algorithm (FPA) (Yang, 2012), Genetic Algorithm (GA) (Mirjalili, 2022), Slap Swarm Algorithm (SSA) (Mirjalili, Gandomi, Mirjalili, Saremi, Faris, & Mirjalili, 2017), Whale Optimization Algorithm (WOA) (Mirjalili & Lewis, 2016) and Harris Hawks Algorithm (HHO) (Heidari, Mirjalili, Faris, Aljarah, Mafarja, & Chen, 2019) in this subsection. Table 5 shows the performance metrics of each algorithm on NSL-KDD dataset.

According to Table 5, the optimal feature subsets produced by the above feature selection algorithms can achieve good results on the test set, and the optimal feature subset selected by using the WOA feature selection algorithm has the best performance on the test set, but the network traffic of NSL-KDD dataset after pre-processing has a total of 41 dimensions, and the optimal feature subset selected by this algorithm is 40, so this method is not feasible. Hence, the bGWO is selected as the feature selection algorithm for the industrial CPS intrusion detection dataset, the algorithm not only reduces the number of features by 49%, but also reduces the accuracy of the Baseline model by only 0.0024, which is basically negligible.

The fitness curves of each algorithm are as illustrated in Fig. 8, respectively. From the following figures, we can see that the fitness of the optimal feature subset obtained using the bGWO algorithm (please refer to Equation (15) for the fitness formula) is the best, which is 0.014, and it also verifies that the bGWO algorithm has better feature selection

Table 3
Baseline of NSL-KDD.

Model	Accuracy	Precision	DR	FMeasure	Parameters	Model size
Teacher	0.9884	0.9886	0.9884	0.9885	33,589	131.2 KB
Student	0.9686	0.9671	0.9686	0.9668	4,624	18.1 KB

Table 4

Baseline of CIC IDS2017.

Model	Accuracy	Precision	DR	FMeasure	Parameters	Model size
Teacher	0.9952	0.9956	0.9954	0.9951	33,589	131.2 KB
Student	0.9797	0.9802	0.9799	0.9761	4,624	18.1 KB

Table 5

Performance of each algorithm.

Algorithm	Accuracy	Precision	DR	FMeasure	Feature size
bGWO	0.9860	0.9869	0.9860	0.9863	21
FPA	0.9846	0.9852	0.9846	0.9848	21
GA	0.9802	0.9806	0.9802	0.9802	22
SSA	0.9850	0.9850	0.9850	0.9848	23
WOA	0.9894	0.9901	0.9894	0.9897	40
HHO	0.9841	0.9846	0.9841	0.9843	26

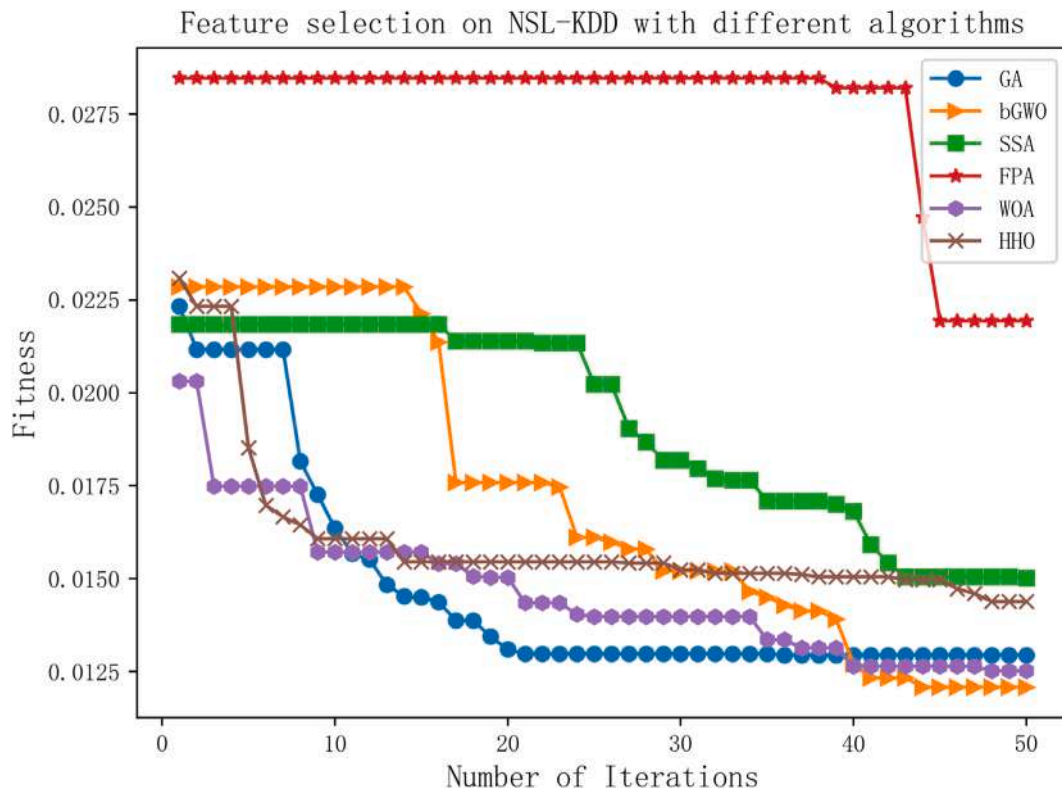
capability. Consequently, bGWO is also used as the feature selection algorithm on the CIC IDS2017 dataset, so we will not do more validation here.

6.2. Classification results of NSL-KDD and CIC IDS2017 datasets

In order to verify the superiority of the KD-TCNN compared with the Baseline model, we will train the KD-TCNN model on the industrial CPS intrusion detection benchmark datasets and compare it with the Baseline model, and the comparison results are shown in Fig. 9. It can be seen in Fig. 9(a) that since the Teacher model and the Student model are obtained using NSL-KDD dataset which has just been pre-processed as well as has not been subjected to any further treatments, what's more, the Teacher model has 7 times the number of parameters than that of the Student model and KD-TCNN model. It seems to reason that the Teacher model's performance metrics are superior to those of the Student model and the KD-TCNN model. However, the KD-TCNN model does not differ

much from the Teacher model in all performance metrics. In terms of accuracy, the KD-TCNN model is only 0.4% lower than the Teacher model, while it is 1.58% higher than the Student model. The KD-TCNN model is only 0.34% worse than the Teacher model, while it is 1.83% higher than the Student model in FMeasure. From Fig. 9(b), we can see that the KD-TCNN model is almost the same as the Teacher model in all performance metrics. In terms of accuracy, the KD-TCNN model is only 0.08% lower than the Teacher model while it improves 1.46% relative to the Student model, and with regards to detection rate, the KD-TCNN model is only 0.07% lower than the Teacher model, while it is 1.48% greater than the Student model. This is mainly because: (1) The outputs of the student network are pushed as close as possible to the teacher network through knowledge distillation, which can allow the student network to gain the categorical knowledge from the teacher network, thus improving the student model's recognition of network traffic categories with few samples. (2) Knowledge distillation can utilize the ability of deep metric learning to reduce the difference between comparable inputs to lower the gap between teacher and student model outputs, hence boosting the student model's performance. (3) Using the idea of pre-training and fine-tuning, the model's performance can be improved greatly via K-fold cross training. Therefore, we can draw the conclusion that the proposed knowledge distillation model based on triplet convolution neural network can successfully minimize the number of model parameters while also enhancing the performance metrics of the tiny model on intrusion detection datasets at the same time.

We conducted comparison experiments using traditional deep learning algorithms such as deep neural network (DNN), convolution neural network (CNN) and recurrent neural network (RNN) to verify the

**Fig. 8.** Fitness curves of each feature selection algorithm.

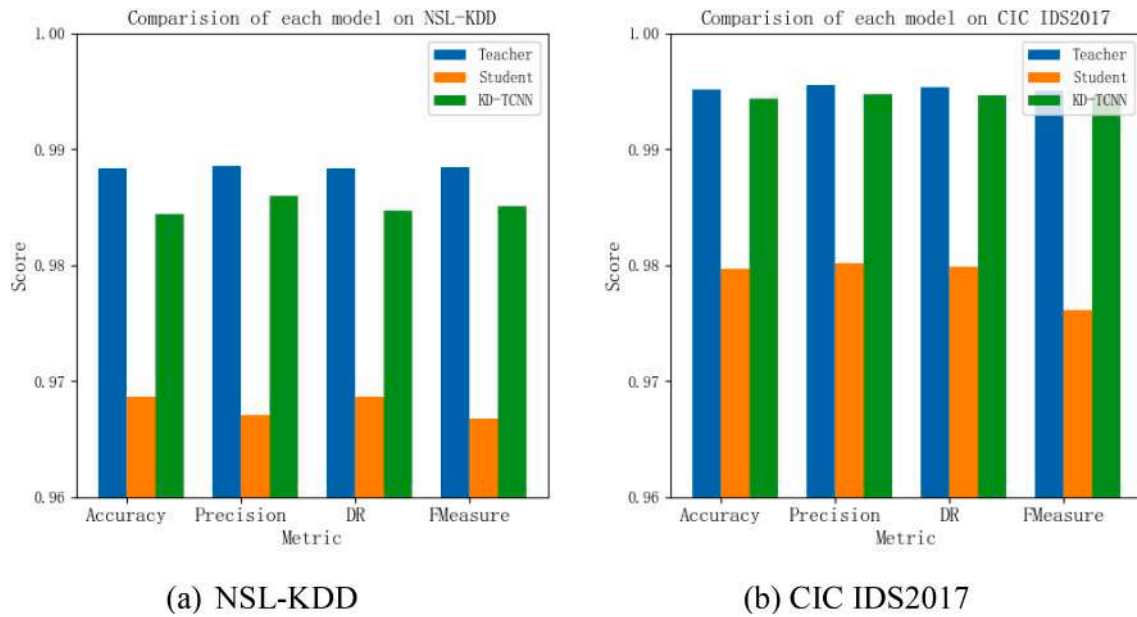


Fig. 9. Performance comparison of each model.

validity of the KD-TCNN model in our paper. In order to make the experiment more rigorous, we will use DNN, CNN and RNN with similar structure to the proposed KD-TCNN model. Tables 6 and 7 show the experimental results of multi-classification for the NSL-KDD and CIC IDS2017 datasets using these traditional deep learning models and the KD-TCNN model, respectively. Although the SVMSMOTE algorithm mentioned in Section 4.2 alleviates the data imbalance problem, the intrusion detection datasets still have slight data imbalance issues, if we only use accuracy as the performance indicator, which cannot reflect the models' ability to classify abnormal traffic in the unbalanced dataset, hence detection rate, precision and FMeasure are also used to evaluate the multi-classification performance of the models.

The results, reported in Table 6 for all the models, show that the KD-TCNN model has the best multi-classification detection results on the NSL-KDD dataset, which can achieve the accuracy of 98.44%, what's more, the rest of the metrics can also reach above 98%, which is mainly because the KD-TCNN model can fully explore the connection between each feature through deep metric learning and learn a better feature representation. Although the detection effect of CNN and KD-TCNN models for multi-classification is very close, the number of parameters of KD-TCNN model is only 19.9% of CNN model, and the model size is 72.5 KB smaller than CNN model. Besides, the KD-TCNN model is better than the DNN and RNN models in each evaluation metric, and in terms of accuracy, the KD-TCNN model is 4.42% and 3.75% greater than the DNN and RNN models, respectively. From the FMeasure's point of view, the KD-TCNN model is 5.69% and 4.85% higher than DNN and RNN models, respectively, while the number of KD-TCNN model parameters only accounts for 91.4% and 24.1% of DNN and RNN models. We can see from Table 7 that the multi-classification detection results on the CIC IDS2017 dataset used KD-TCNN model is similar to the NSL-KDD dataset, and regarding to accuracy, the KD-TCNN model is 0.63%, 0.05% and 0.54% greater than the DNN, CNN and RNN models, respectively. In

terms of accuracy, the KD-TCNN model is 0.98%, 0.06% and 0.57% higher than the DNN, CNN and RNN models, respectively. Overall, in the intrusion detection process of the NSL-KDD and CIC IDS2017 datasets, each method can achieve an outstanding result for each performance metric, indicating that DNN, CNN, RNN as well as KD-TCNN model can successfully identify abnormal network intrusion traffic, however, our model's multi-classification results are much better than traditional deep learning models in most cases. Since the knowledge distillation framework is applied to the KD-TCNN model, not only the number of parameters and the model size of the KD-TCNN model are significantly reduced compared with DNN, CNN and RNN, but also the KD-TCNN model still has the best multi-classification detection effect on the industrial CPS intrusion detection datasets.

The confusion matrix, referred as the most basic, obvious, and straightforward approach of assessing a model's performance, is also used in our paper. It uses heat map, color difference and brightness to show the difference of data. It allows the records in the datasets to be aggregated according to the real and predicted results, furthermore, the confusion matrix can also be used to obtain the model assessment metrics. The confusion matrices of the results for the multi-classification experiments of each model on the NSL-KDD and CIC IDS2017 datasets are shown in Figs. 10 and 11. From Figs. 10 and 11, we can conclude that the KD-TCNN model has the best classification effect. When the confusion matrix has the largest value and the darkest color on the diagonal, which means that most of the sample categories can be successfully predicted. In NSL-KDD dataset, the DNN model predicted 96% of R2L and 84% of U2R as Normal, respectively, the RNN model predicted 86% of R2L as Normal, and the CNN model also predicted 38% of R2L as Normal. On the CIC IDS2017 dataset, the DNN model predicted 41% of Slowhttptest and 83% of Slowloris as Benign, respectively, while the RNN model also predicted 32% of Slowhttptest and 71% of Slowloris as Benign, and the CNN model predicted 43% of Slowhttptest predicted as

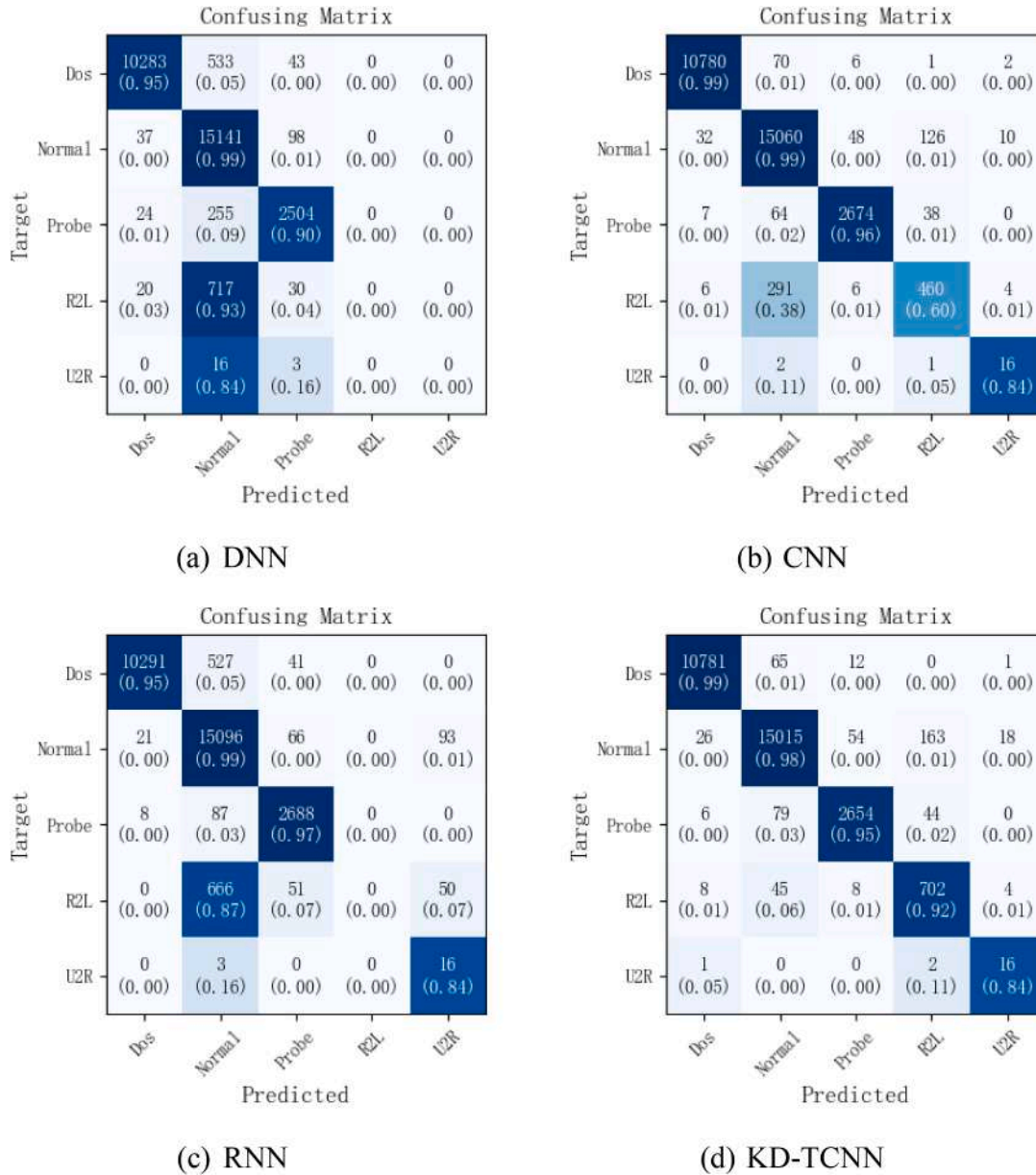
Table 6
Multi-classification results of NSL-KDD dataset.

Model	Accuracy	Precision	DR	FMeasure	Parameters	Model size
DNN	0.9402	0.9183	0.9405	0.9282	5,061	19.8 KB
CNN	0.9766	0.9759	0.9769	0.9761	23,184	90.6 KB
RNN	0.9469	0.9279	0.9472	0.9366	19,200	75 KB
KD-TCNN	0.9844	0.9860	0.9847	0.9851	4,624	18.1 KB

Table 7

Multi-classification results of CIC IDS2017 dataset.

Model	Accuracy	Precision	DR	FMeasure	Parameters	Model size
DNN	0.9881	0.9850	0.9884	0.9864	5061	19.8 KB
CNN	0.9939	0.9942	0.9941	0.9938	23,184	90.6 KB
RNN	0.9890	0.9891	0.9892	0.9875	19,200	75 KB
KD-TCNN	0.9944	0.9948	0.9947	0.9946	4,624	18.1 KB

**Fig. 10.** Multi-classification confusion matrix for each model on NSL-KDD.

Benign. At this time, the model has a plenty of false positives as well as false negatives, which undoubtedly greatly reduces the accuracy of the IDS and increases the possibility of anomalies on IoT devices in the industrial CPS, which also illustrates the effectiveness of the knowledge distillation intrusion detection model based on the triplet convolution neural network proposed in our paper.

For the sake of further demonstrating the superiority of the KD-TCNN model proposed in our paper, the recent state-of-the-art models on NSL-KDD and CIC IDS2017 datasets are compared with our KD-TCNN model as well, Tables 8 and 9 below give the experimental results, where “-”

donates that the results of the performance metric are not presented in these papers. From Table 8, we can see that the KD-TCNN model is only lower than the LOGNN model in the matter of accuracy and detection rate compared with the recent state-of-the-art models, which is mainly because the LOGNN model uses a larger network structure and has more parameters than our model, while the KD-TCNN model is a lightweight model and the number of parameters it has is the trade-off between detection speed and efficiency. However, the results obtained by the KD-TCNN model still outperform those state-of-the-art approaches, even greater than the IGAN-IDS model by 13.99% in accuracy, and higher

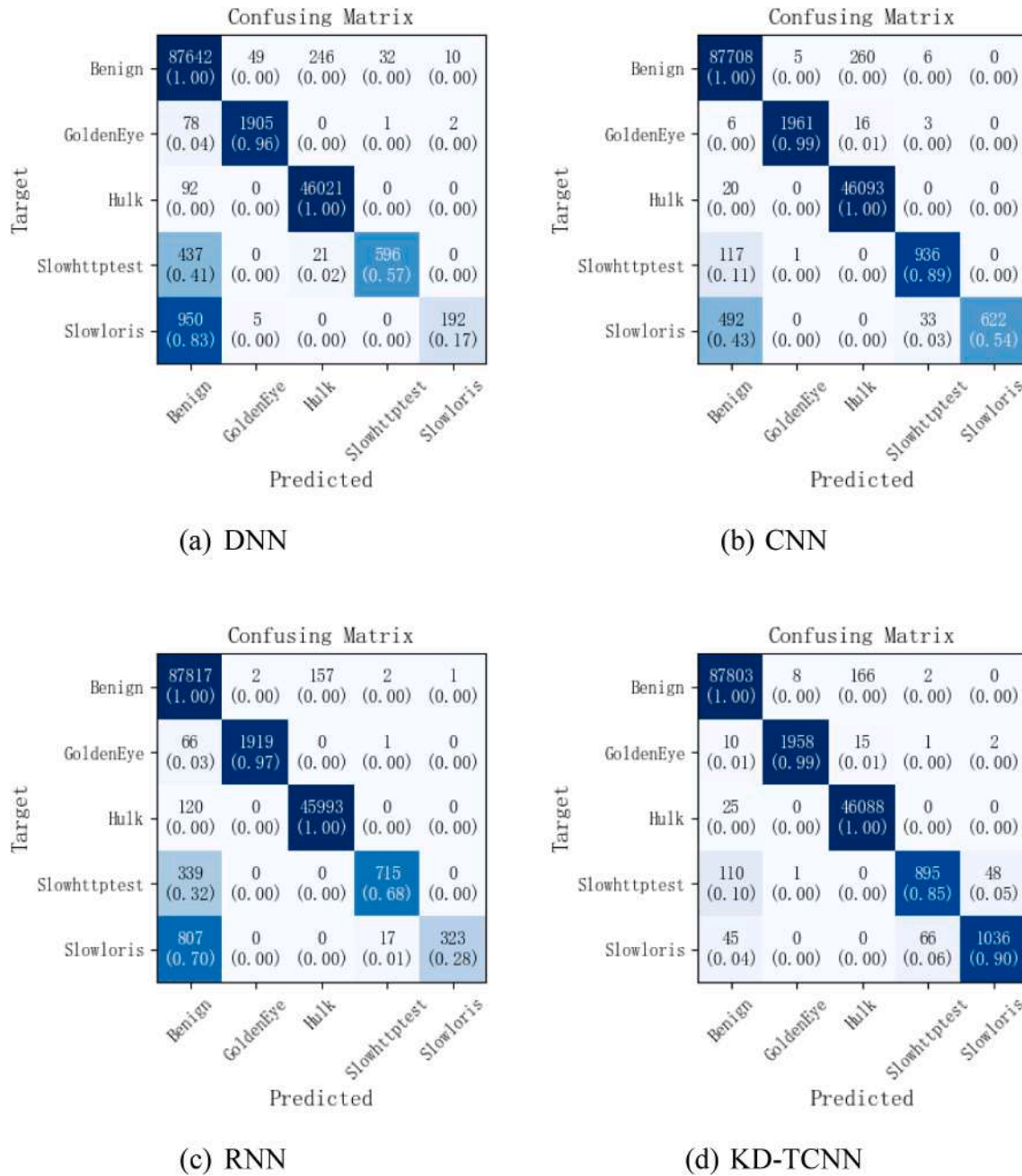


Fig. 11. Multi-classification confusion matrix for each model on CIC IDS2017.

Table 8

Multi-classification results of NSL-KDD dataset.

Model	Accuracy	Precision	DR	FMeasure
OCNN-HMLSTM (Rajesh Kanna & Santhi, 21071, 2021)	0.9067	0.8671	0.9519	0.9146
LOGNN (Wang, Xu, & He, 2021)	0.9870	0.9840	0.9870	0.9850
IGAN-IDS (Huang & Lei, 1021, 2020)	0.8445	–	–	0.8417
PSO-DBN (Elmasry, Akbulut, & Zaim, 2020)	0.9691	0.9810	0.9229	0.9511
KD-TCNN	0.9844	0.9860	0.9847	0.9851

than the OCNN-HMLSTM and PSO-IDS models by 7.77% and 1.53%. With regard to FMeasure, the result obtained by the KD-TCNN model is 98.51%, which is greater than LOGNN, OCNN-HMKSTM, IGAN-IDS and PSO-DBN by 0.01%, 7.05%, 14.34% and 3.4% respectively, which

Table 9

Multi-classification results of CIC IDS2017 dataset.

Model	Accuracy	Precision	DR	FMeasure
PSO-DBN (Elmasry et al., 2020)	0.9581	0.9582	0.9581	0.9581
SVM-GAC (Shukla, 2021)	0.9993	0.9873	0.9961	0.9891
NSGA2-LR (Khammassi & Krichen, 1071, 2020)	0.9516	–	0.8532	–
CCFS-DT (Farahani, 2020)	0.9791	0.9942	0.9907	0.9924
KD-TCNN	0.9944	0.9948	0.9947	0.9946

indicates that our KD-TCNN model can effectively detect and classify network traffic of various attack categories for network intrusion traffic. As shown in Table 9, as to accuracy, the KD-TCNN model is only 0.49% lower than the SVM-GAC model, while it is 3.63%, 4.28% and 1.53% greater than the PSO-DBN, NSGA2-LR and CCFS-DT model, respectively,

and with regard to detection rate, the KD-TCNN model is solely 0.14% lower than the SVM-GAC model, while it is 3.66%, 14.15% and 0.4% higher than the PSO-DBN, NSGA2-LR, and CCFS-DT models by 3.66%, 14.15%, and 0.4%, respectively. In a word, each model mentioned above can still obtain a superior result for each performance metric in the CIC IDS2017 dataset's intrusion detection procedure, mainly because the CIC IDS2017 dataset is simpler than NSL-KDD dataset and has more sufficient network traffic.

Each layer of the convolution neural network can extract fine-grained different traffic features to aid the convolution neural network in understanding the classification of data as "benign" or "abnormal" and the specific classification of anomalies into different abnormal classes during the intrusion detection process. The last layer of the convolution neural network, named global pooling layer, has the richest features. In order to have a better intuitive understanding of this process, we randomly selected 20,000 network flows into the KD-TCNN model and used the t-SNE algorithm (van der Maaten & Hinton, 2008) to downscale and visualize the global pooling layer of the KD-TCNN model and the original dataset. Fig. 12(a) and (b) show the original dataset of NSL-KDD and the visualized images after processing by the KD-TCNN model, respectively, and Fig. 13(a) and (b) show the original dataset of CIC IDS2017 and the visualized images after processing by the KD-TCNN model, respectively. From Fig. 12(a) and Fig. 13(a), we can see that the data of the NSL-KDD and CIC IDS2017 datasets without processing by the KD-TCNN model cannot be linearly separable, and the Probe attacks are scattered in clusters of Dos and Normal categories and GoldenEye attacks are scattered in clusters of Benign and Hulk categories, which means that the Probe attacks are easily classified by the classifier into Dos and Normal categories, while GoldenEye attacks are easily classified by the classifier into Benign and Hulk categories, which makes the classifier model unreliable and leads to failure in intrusion detection of attack traffic. From Fig. 12(b) and Fig. 13(b), comparing the visualized images of anomaly detection obtained without processing by the KD-TCNN model, it can be seen that the different categorical network traffic shows an aggregation trend in the low dimensional space, which can also demonstrate that the network traffic after the student network processing has linear separability and can be used to accurately identify different kinds of anomaly network traffic.

6.3. Ablation study

In order to evaluate the validity of the four parts of the KD-TCNN model proposed in this paper, we will undertake ablation studies on the KD-TCNN model. The proposed model contains four main

components, which are feature selection, deep metric learning, knowledge distillation and K-fold cross training, so we conducted ablation experiments on the NSL-KDD dataset for each of these four components, and Table 10 shows the ablation experimental results. It can be seen from the Table 10 that the accuracy of the original model is the lowest, which is 96.86%, after adding the feature selection operation to eliminate redundant features and improve the performance of the model, the accuracy of the model is 96.88% at this time. We then introduced knowledge distillation into our model, and then the accuracy of the model improved by 0.39% compared to the original model. After introducing deep metric learning into the knowledge distillation framework to reduce the difference between the teacher network's outputs and the student network's outputs, the model's accuracy achieves to 97.98%, and then introducing the K-fold cross training method to train the model further increased the accuracy of the model to 98.44%. Compared with the results of Teacher model in the Baseline, the accuracy decreased only 0.4%, which fully proves the effectiveness of the knowledge distillation intrusion detection model based on triplet convolution neural network and the K-fold cross training method proposed in this paper.

7. Conclusion and future work

In our paper, we proposed the KD-TCNN model, a triplet convolution neural network-based knowledge distillation model for industrial CPS intrusion detection. Knowledge distillation is utilized in this paper to bring the outputs of the student network as close to that of teacher network as possible, which can make the student network to learn the information between categories in the teacher network, and can process and analyze large-scale data as well as minimize the number of model parameters. Our approach use deep metric learning to decrease the distance between similar inputs for knowledge distillation to lower the gap between the teacher network and the student network outputs and improve the performance of the student model, and deep separable convolution is applied to further reduce the computational resource as well as the number of model parameters, so that the intrusion detection model can be deployed to nodes with limited computational power in the IoT network to reduce the intrusion detection time so that real-time detection can be achieved. The KD-TCNN model is validated on the industrial CPS intrusion detection benchmark dataset NSL-KDD and CIC IDS2017, respectively. The experimental results show that the KD-TCNN model is better than the traditional deep learning models in terms of both parametric size and other performance metrics.

The KD-TCNN model proposed in our paper is still able to achieve

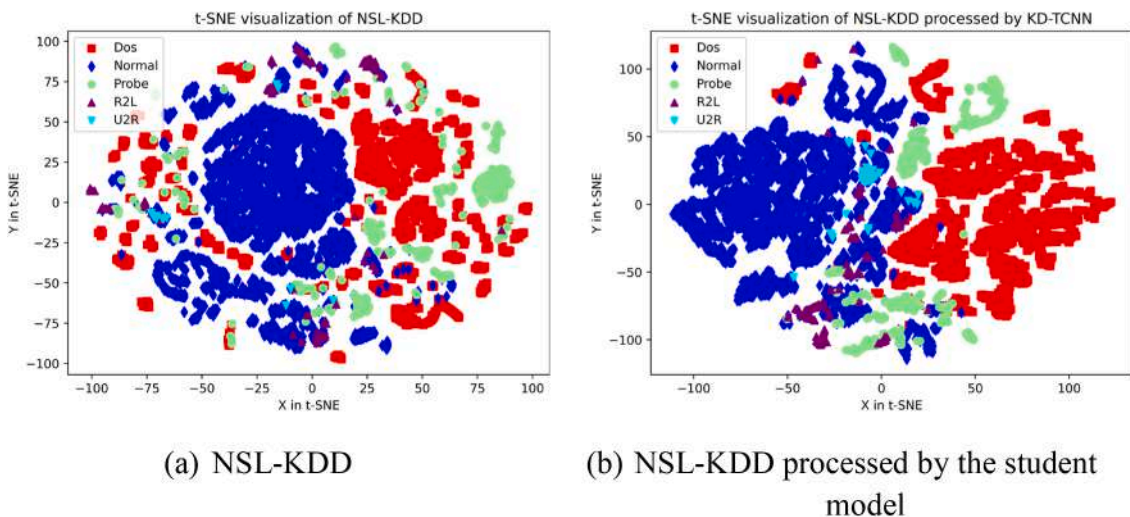


Fig. 12. t-SNE visualization of NSL-KDD.

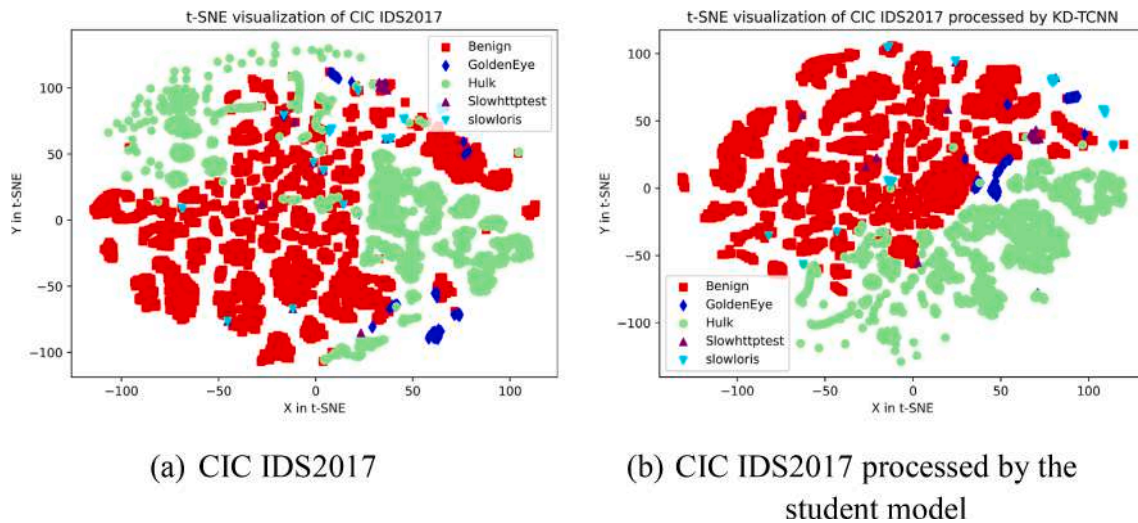


Fig. 13. t-SNE visualization of CIC IDS2017.

Table 10
Ablation study on NSL-KDD.

Feature selection	Deep metric learning	Knowledge distillation	K-fold cross training	Accuracy
×	×	×	×	0.9686
✓	×	×	×	0.9688
✓	×	✓	×	0.9725
✓	✓	✓	×	0.9798
✓	✓	✓	✓	0.9844

excellent anomaly detection results when compared to recent state-of-the-art intrusion detection models. However, the KD-TCNN model has more hyperparameters to be optimized, and the hyperparameters have a large impact on the KD-TCNN model. Using manual verification of the hyperparameters is time-consuming, and the Hyperband algorithm (Lisha Li, Kevin Jamieson, Giulia DeSalvo, Afshin Rostamizadeh, Ameet Talwalkar. "Hyperband: Bandit-Based Configuration Evaluation for Hyperparameter Optimization," *International Conference on Learning Representations*, 2017, 2017) can be considered subsequently for hyperparameter tuning. Moreover, the KD-TCNN model requires two steps (1. teacher model pre-training; 2. KD-TCNN model training) to implement the intrusion detection process. The performance of the KD-TCNN model depends largely on the goodness of the pre-trained teacher model, while subsequent teacher model and student model can be considered to be trained simultaneously to reduce the dependence of the student model on the teacher model as well as improve the efficiency of the model.

CRediT authorship contribution statement

Zhendong Wang: Conceptualization, Data curation, Formal analysis, Resources, Supervision, Project administration, Writing – review & editing. **Zeyu Li:** Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft, Writing – review & editing. **Daojing He:** Conceptualization, Writing – review & editing. **Sammy Chan:** Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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